Memorandum of Understanding — UNIDO × SCUT (optimized draft)

联合国工业发展组织与华南理工大学谅解备忘录（优化稿）

Memorandum of Understanding between the United Nations Industrial Development Organization and South China University of Technology

PART I — ENGLISH TEXT (for signature)

Memorandum of Understanding

between

the United Nations Industrial Development Organization (UNIDO)

and

South China University of Technology (SCUT)

THE UNITED NATIONS INDUSTRIAL DEVELOPMENT ORGANIZATION (“UNIDO”), an international organization established by the Constitution of UNIDO, with its Headquarters at Vienna International Centre, P.O. Box 300, 1400 Vienna, Austria, represented by [Name], [Title],

and

SOUTH CHINA UNIVERSITY OF TECHNOLOGY (“SCUT”), a national university of the People's Republic of China, with its address at No. 381 Wushan Road, Tianhe District, Guangzhou, China, represented by [Name], President,

hereinafter referred to individually as “a Party” and collectively as “the Parties”,

Recalling UNIDO's mandate to promote inclusive and sustainable industrial development and to contribute to the achievement of the 2030 Agenda for Sustainable Development, in particular Sustainable Development Goals 9, 12, 13 and 17;

Recognizing SCUT's research, engineering and testing capacity in pulp and paper, plant-fiber materials and paper-based substitutes for plastics, anchored in the State Key Laboratory of Advanced Papermaking and Paper-based Materials and the National Engineering Research Center for Paper and Pollution Control, and extending, through partnerships with pulp, paper and packaging enterprises, to pilot-scale production and industrial validation of nanocellulose and functional paper-based materials;

Noting the growing demand from member States and from producers in developing countries for harmonized and verifiable methodologies to measure and substantiate the substitution of plastics by plant-fiber materials,

building on UNIDO's ongoing work on environmental footprint guidelines;

Desiring to cooperate in the development of methodologies, conformity tools and technical public goods, and in their demonstration at pilot and industrial scale, in support of fair production, sustainability standards and trade;

Have agreed as follows:

Article 1 — Purpose

The purpose of this Memorandum of Understanding is to establish a framework of cooperation between UNIDO and SCUT in the field of plant-fiber materials for plastic substitution, with a view to developing and disseminating methodologies and conformity tools that are usable by member States and by producers in developing countries, and to contribute technical inputs to the work of the relevant standardization bodies.

This MoU is a framework instrument. Specific activities shall be implemented through work plans or project documents agreed in writing by the Parties. For the purposes of this MoU, “plant-fiber materials” means materials derived wholly or predominantly from plant fibres, including paper-based and nanocellulosic products, intended to substitute for plastics in defined applications. “The Network” means the international academic–industry cooperation network of participating institutions referred to in Article 6, which has no legal personality.

Article 2 — Areas of cooperation

The Parties shall cooperate, within their respective mandates and available resources, in the following areas:

1. Carbon footprint methodology. Development of a methodology for calculating the carbon footprint of paper-based and plant-fiber substitutes for plastics, addressing biogenic carbon accounting, avoided-emissions boundaries and end-of-life treatment, together with an auditable reporting template for exporting firms.

2. Degradability and recyclability assessment. Development of assessment methods for degradability, recyclability and material performance of plant-fiber substitutes, including barrier properties and food-contact safety, with a view to improving the comparability of test results across markets.

3. Technology–industry mapping. Preparation of a global mapping of plant-fiber substitutes for plastics, distinguishing commercialized, pilot-stage and frontier technologies, with annotations on performance boundaries, cost ranges, applicable scenarios and indicative abatement potential, drawing on production and pilot data from SCUT and its industrial partners.

4. South–South technical dialogue. Convening, under UNIDO's convening function, of an international technical forum bringing together policy makers, industry leaders and academic experts from interested member States.

5. Capacity building and training. Design and delivery of modular training and technology packages for enterprises and institutions in developing countries.

6. Joint research and scientific exchange. Joint research activities, exchange of experts, joint publications and participation in scientific events in the areas above.

7. Industrial validation and demonstration. Pilot-scale validation of the methodologies under items 1 and 2, drawing on production data and demonstration lines of SCUT and its industrial partners, to support technology uptake by producers in developing countries.

Article 3 — Forms of cooperation

Cooperation under this MoU may take the following forms: joint studies and methodology development; joint workshops, seminars and conferences; exchange of experts and visiting researchers; joint publications and dissemination of technical materials; use of laboratory, pilot and testing facilities where agreed; and such other forms as the Parties may agree in writing.

Article 4 — Responsibilities of UNIDO

UNIDO shall, within its mandate and subject to its rules and procedures:

1. provide technical guidance on the alignment of the activities with the UNIDO service package on fair production, sustainability standards and trade;
2. participate in the review of methodology drafts and technical outputs;
3. convene, where appropriate, the South–South technical dialogue and facilitate the participation of member States;
4. disseminate the public goods produced under this MoU through its networks and platforms;
5. designate a technical focal point for the coordination of activities under this MoU.

Nothing in this MoU shall be construed as creating a financial obligation on the part of UNIDO.

Article 5 — Responsibilities of SCUT

SCUT shall:

1. act as legal host and operational anchor of the cooperation;
2. host the secretariat of the Network within the State Key Laboratory of Advanced Papermaking and Paper-based Materials, and provide the necessary office space, personnel and administrative support;
3. lead the technical development of the methodology package and the technology–industry mapping, and provide access to laboratory, pilot and testing facilities;
4. mobilize industrial partners for data provision, pilot validation and user feedback;
5. bear the costs of its own participation, including staff, facilities and travel, unless otherwise agreed in writing;
6. manage conflicts of interest arising from commercial interests of its staff in enterprises active in this field, in accordance with Article 6, paragraph 5;
7. designate a technical focal point for the coordination of activities under this MoU.

Article 6 — Governance and coordination

1. A Joint Steering Committee shall be established, co-chaired by a representative of UNIDO and a representative of SCUT, to approve annual work plans, review progress and decide on adjustments. It shall meet [at least once a year], in person or by video conference, and shall take decisions by consensus.

2. A secretariat, hosted by SCUT at the State Key Laboratory, shall be responsible for day-to-day coordination, documentation and reporting.

3. The Parties shall exchange progress reports [every six months].

4. The Network may be described publicly as an international academic–industry cooperation network hosted by SCUT and cooperating with UNIDO under this MoU. It shall not use the name or emblem of UNIDO in its title, and shall not be presented as a UNIDO office, subsidiary body or legally separate entity.

5. Independence of review. The review of methodology drafts shall be chaired by an independent testing or certification body acceptable to both Parties. No enterprise in which SCUT staff hold a commercial interest shall take part in the review of a method that it would itself apply.

Article 7 — Financial arrangements

Each Party shall bear its own costs arising from the implementation of this MoU. Activities requiring funding shall be subject to separate written arrangements and to the respective financial regulations and procedures of the Parties. No funds shall be transferred between the Parties under this MoU, unless otherwise agreed in writing, including through project documents or technical cooperation arrangements.

Article 8 — Intellectual property

1. Background intellectual property shall remain the property of the Party that owns it.

2. Intellectual property arising from joint activities shall be owned [jointly by the Parties / as specified in the relevant work plan].

3. Methodologies, mappings and training materials produced under this MoU are intended to be disseminated as public goods [under an open licence, such as CC BY 4.0 for texts and a permissive licence for software tools]. Their publication shall be subject to prior consultation between the Parties, to the intellectual property rights of third parties, and to the protection of commercially sensitive information identified by either Party.

4. The name, logo and trademarks of either Party shall not be used in publications or promotional materials without prior written consent.

Article 9 — Use of name and emblem

Neither Party shall use the name, acronym or emblem of the other Party, or any abbreviation thereof, in connection with its business or otherwise, without the express prior written approval of the other Party.

UNIDO's name and emblem are protected under the Convention on the Privileges and Immunities of the Specialized Agencies.

Article 10 — Confidentiality

The Parties shall treat as confidential any information designated as confidential by the disclosing Party, and shall use it only for the purposes of this MoU. This obligation shall not apply to information that is publicly available or independently developed.

Article 11 — Status of personnel

Personnel of one Party shall not be considered as staff members of the other Party. They shall remain subject to the respective rules and regulations of their own institution. Nothing in this MoU shall be construed as conferring on either Party's personnel any privileges or immunities, or as affecting the privileges and immunities enjoyed by UNIDO under the Convention on the Privileges and Immunities of the Specialized Agencies.

Article 12 — Duration, amendment and termination

1. This MoU shall remain in force for [three (3)] years from the date of its signature, and may be renewed by written agreement of the Parties.
2. It may be amended by written agreement of the Parties.
3. Either Party may terminate this MoU by giving [three (3)] months' written notice to the other Party. Termination shall not affect the completion of activities already agreed in writing.

Article 13 — Settlement of disputes

Any dispute arising from the interpretation or implementation of this MoU shall be settled amicably through consultation and negotiation between the Parties.

Article 14 — Entry into force

This MoU shall enter into force upon signature by the duly authorized representatives of both Parties. Except for Articles 8, 9, 10 and 11, which are intended to be binding, this MoU does not create legally binding obligations; specific activities shall be governed by separate written arrangements.

Done in [two (2)] originals in the English and Chinese languages, [the English text prevailing / both texts being equally authentic].

Signature

For the United Nations Industrial Development Organization

Name: _____ Title: _____

Signature: _____ Date: _____

For South China University of Technology

Name: _____ Title: _____

Signature: _____ Date: _____

Annex I — Indicative work plan (18 months)

Period	Activities	Outputs	Lead
Months 1–3	Technical alignment; scope of methodology confirmed; work plan agreed	Agreed work plan; designated focal points	UNIDO / SCUT
Months 4–6	Methodology framework; data collection protocol; mapping methodology	Framework document; protocol	SCUT
Months 7–9	Methodology Draft V0.1 validated against production data; mapping data collection	Draft V0.1; workbook (beta)	SCUT with industrial partners
Months 10–12	Independent review of Draft V0.1; mapping v1 published; training module drafted	Review report; mapping v1	UNIDO / SCUT
Months 13–18	Methodology V1.0; South–South technical dialogue; pilot training in one region	Methodology V1.0; forum report; training package	SCUT / UNIDO

PART II — 中文译本（供校内审议参考）

中华人民共和国华南理工大学与联合国工业发展组织谅解备忘录

联合国工业发展组织（以下简称“UNIDO”），系依据《联合国工业发展组织章程》设立的国际组织，总部设在奥地利维也纳维也纳国际中心，由[姓名]、[职务]代表；

与

华南理工大学（以下简称“SCUT”），系中华人民共和国的一所全国重点大学，地址为中国广州市天河区五山路381号，由[姓名]、校长代表；

（单称“一方”，合称“双方”）

忆及 UNIDO 促进包容与可持续工业发展、推动落实2030年可持续发展议程尤其是可持续发展目标9、12、13、17的职责；

认识到 SCUT 在制浆造纸、植物纤维材料及纸基替代塑料材料领域的研究、工程化与检测能力，以先进造纸与纸基材料全国重点实验室和造纸与污染控制国家工程研究中心为依托，并通过与造纸、包装企业的产业伙伴关系，延伸至纳米纤维素与功能纸基材料的中试放大与产业化验证；

注意到成员国及发展中国家生产商对统一、可核查的植物纤维材料替代塑料计量方法的需求日益增长，并愿依托 UNIDO 环境足迹指南工作的既有成果推进相关方法开发；

愿在支持公平生产、可持续标准与贸易的方法学、合规工具和技术公共产品及其小试、中试与产业化示范方面开展合作；

达成如下谅解：

第一条　宗旨

本谅解备忘录旨在建立 UNIDO 与 SCUT 在植物纤维材料替代塑料领域的合作框架，共同开发并推广成员国及发展中国家生产企业可用的方法学与合规工具，并为相关标准化机构的工作提供技术输入。

本备忘录为框架性文件，具体活动通过双方书面同意的工作计划或项目文件实施。本备忘录所称“植物纤维材料”，指全部或主要由植物纤维制成、用于在特定应用中替代塑料的材料，包括纸基材料与纳米纤维素产品。本备忘录所称“合作网络”，指第六条所述由参与机构组成的国际学术—产业合作网络，不具有法人资格。

第二条　合作领域

双方在各自职责与可用资源范围内，在以下领域开展合作：

1. 碳足迹方法学：开发纸基及植物纤维替代塑料产品的碳足迹核算方法，涵盖生物碳核算、避免排放边界与末端处置，并配套面向出口企业的可审计报告模板。

2. 降解性与可回收性评价：开发降解性、可回收性及材料性能评价方法，包括阻隔性能与食品接触安全，以提高不同市场间检测结果的可比性。

3. 技术—产业图谱：编制全球植物纤维材料替代塑料的技术-产业图谱，区分成熟技术、中试阶段与前沿研究，标注性能边界、成本区间、适用场景与减排潜力，并纳入 SCUT 及其产业伙伴的生产与中试数据。

4. 南南技术对话：在 UNIDO 的召集职能下举办国际技术论坛，汇集相关成员国政策制定者、产业领袖与学术专家。

5. 能力建设与培训：为发展中国家企业和机构设计并实施模块化培训与技术包。

6. 联合研究与学术交流：在上述领域开展联合研究、专家交流、联合发表及学术活动。

7. 产业化验证与示范：依托 SCUT 及其产业伙伴的生产数据与示范线，对第1、2项所开发的方法开展中试与产业化验证，支持发展中国家生产商的技术应用。

第三条　合作形式

合作可采取以下形式：联合研究与方法学开发；联合举办研讨会、讲习班与会议；专家与访问学者交流；联合发表与传播技术材料；经同意使用实验室、中试与检测设施；以及双方书面约定的其他形式。

第四条　UNIDO 的责任

UNIDO 在其职责及规则程序范围内：

1. 就活动与 UNIDO 公平生产、可持续标准与贸易服务包的衔接提供技术指导；

2. 参与方法学草案与技术成果的评审；

3. 视情召集南南技术对话，并为成员国参与提供便利；

4. 通过其网络与平台传播本备忘录项下形成的公共产品；

5. 指定技术对接人，协调本备忘录项下活动。

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第五条　SCUT 的责任

SCUT：

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2. 在先进造纸与纸基材料全国重点实验室设立合作网络秘书处，提供必要办公场所、人员与行政支持；
3. 牵头方法学包与技术-产业图谱的技术开发，并提供实验室、中试与检测设施；
4. 组织产业伙伴提供数据、开展中试验证并反馈使用意见；
5. 除另有书面约定外，自行承担本方人员、设施与差旅费用；
6. 对本校人员在相关企业中持有商业利益所产生的利益冲突进行管理，具体按第六条第5款执行；
7. 指定技术对接人，协调本备忘录项下活动。

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1. 设立联合指导委员会，由 UNIDO 代表与 SCUT 代表共同担任主席，负责批准年度工作计划、审议进展并决定调整事项；每年至少召开一次会议，可线上举行，并以协商一致方式作出决定。
2. 设立秘书处，由 SCUT 依托全国重点实验室运行，负责日常协调、文档与报告。
3. 双方每六个月交换一次进展报告。
4. 合作网络对外可表述为“由华南理工大学主办、依据本备忘录与 UNIDO 开展合作的国际学术—产业合作网络”；网络名称不使用 UNIDO 名称与徽标，且不得表述为 UNIDO 的办公室、附属机构或独立法律实体。
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1. 本备忘录自签署之日起有效期[三（3）]年，经双方书面同意可续期。

2. 经双方书面同意可对本备忘录进行修改。

3. 任何一方可提前[三（3）]个月书面通知终止本备忘录；终止不影响已书面约定的活动的完成。

第十三条　争议解决

因本备忘录的解释或履行产生的争议，由双方通过友好协商解决。

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本备忘录以英文和中文签署[两（2）]份正本，[以英文文本为准／两种文本同等作准]。

签署

联合国工业发展组织

姓名：_____　职务：_____

签字：_____　日期：_____

华南理工大学

姓名：_____　职务：_____

签字：_____　日期：_____

附件一　指示性工作计划（18个月）

时段	活动	产出	牵头方
第1–3月	技术对接；确认方法学范围；商定工作计划	工作计划；指定对接人	UNIDO / SCUT
第4–6月	方法学框架；数据采集规程；图谱方法	框架文件；数据规程	SCUT
第7–9月	以生产数据验证方法学草案 V0.1；图谱数据采集	草案 V0.1；计算工作簿（试用版）	SCUT 及产业伙伴
第10–12月	独立评审草案 V0.1；发布图谱第一版；编制培训模块	评审报告；图谱 v1	UNIDO / SCUT
第13–18月	发布方法学 V1.0；举办南南技术对话；在一个区域试点培训	方法学 V1.0；论坛报告；培训包	SCUT / UNIDO